

Correcting Survey Bias in Childhood Immunization Coverage Estimates Using Bayesian Post-Stratification

A Simulation-Based Study

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Abstract

National statistical systems in developing contexts frequently grapple with non-ignorable non-response bias and data sparsity, particularly when surveying populations in logistically challenging or conflict-affected regions. This study evaluates the efficacy of Bayesian Multilevel Regression with Post-stratification (MRP) in recovering accurate population parameters under conditions of severe selection bias and structural exclusion. Using a synthetic population of 10 million individuals modeled on Nigerian demographic characteristics, we simulated a vaccination coverage survey ($n=1,018$) characterized by the systematic over-sampling of urban, educated populations and the complete omission of specific rural-north strata. We compared the performance of three estimators: Naive, Classical Post-Stratification, and Bayesian MRP.

Results indicate that while the Naive estimator yielded a substantial upward bias of 6.4 percentage points (relative error: 31.4%), Classical Post-Stratification reduced this error but failed to converge to the truth (bias: 2.5 percentage points) due to its inability to estimate strata with zero observations. In contrast, the Bayesian MRP model, implemented via a non-centered parameterization with JAX acceleration, successfully imputed the missing cells through partial pooling. The model reduced the absolute bias to 1.3 percentage points and produced a 94% Highest Density Interval [0.1929, 0.2415] that encapsulated the true population mean. These findings suggest that MRP offers a robust, cost-effective framework for Small Area Estimation (SAE), enabling National Statistical Offices to generate reliable statistics from imperfect, sparse data in resource-constrained environments.

Keywords: *Bayesian MRP, Non-response Bias, Small Area Estimation, Data Sparsity, Survey Methodology, Nigeria.*

1. Introduction

1.1. Background of the Study

Reliable measurement of childhood immunization coverage is central to public health planning. Immunization surveys inform national strategies, track global progress toward Sustainable Development Goals, and guide resource allocation (UNICEF, 2023). Accurate data are crucial not only for monitoring program performance but also for identifying gaps in equity and access across socioeconomic and geographic groups. Immunization coverage serves as a key indicator of the strength of health systems and has been used by agencies such as WHO, UNICEF, and Gavi to monitor progress toward global vaccine equity targets (WHO, 2024).

Yet, survey-based estimates are vulnerable to nonresponse bias, coverage error, and measurement error—issues particularly pronounced in low- and middle-income countries where logistical constraints and uneven access affect data quality (Shelley & Horner, 2021). In many developing contexts, challenges such as incomplete sampling frames, underreporting, and misclassification of vaccination cards contribute to uncertainty in national estimates (Oliveira et al., 2024). Such sources of bias can distort assessments of immunization program effectiveness and hinder efforts to reach “zero-dose” children in underserved communities.

In Nigeria, for instance, the Demographic and Health Survey (NDHS) and Multiple Indicator Cluster Survey (MICS) are the main sources of vaccination statistics (Wagai et al., 2021). However, both face substantial underrepresentation of rural and less-educated households, leading to systematic bias (Williams et al., 2024; Eudes et al., 2023). Studies have shown that regions with high illiteracy or limited healthcare access tend to have lower survey participation and lower vaccination uptake, resulting in overestimation of national coverage rates (Temitayo Victor Lawal et al., 2023; Ariyibi et al., 2023). These biases can produce misleading estimates of national immunization coverage and obscure inequities across regions or social groups (Mordecai Oweibia et al., 2025). Spatial-temporal analyses have also revealed significant regional disparities in Nigeria’s vaccination rates, with coverage ranging from below 10 percent in some northern states to above 50 percent in parts of the south (Temitayo Victor Lawal et al., 2023).

Traditional weighting or post-stratification methods are often used to correct survey imbalances. However, these classical approaches depend on sufficient representation within each subgroup (e.g., age, education, or region) and can perform poorly when data are sparse (Gelman & Hill, 2007). In cases where certain demographic cells are underrepresented or missing entirely, post-stratified estimates may have large variances or residual bias (Little, 1993). As survey nonresponse grows and populations become more heterogeneous, these techniques may no longer produce reliable estimates of true population coverage.

Recent advances in Bayesian hierarchical modeling offer promising alternatives (McGlothlin & Viele, 2018). Through Multilevel Regression and Post-Stratification (MRP), survey estimates can be adjusted even when certain demographic cells are small or underrepresented. The Bayesian framework allows for partial pooling, borrowing information across similar groups to stabilize estimates and quantify uncertainty more effectively. This approach has been successfully applied to public opinion and health surveys where traditional methods failed to capture population complexity (Downes & Carlin, 2019).

In the context of immunization coverage estimation, Bayesian MRP enables researchers to integrate auxiliary data such as census or administrative records to correct for survey bias and produce small-area or subgroup estimates with credible uncertainty intervals (Hogg et al., 2024). This study therefore applies a Bayesian post-stratification approach to a simulated population mirroring the social and demographic structure of Nigeria, demonstrating how modern statistical tools can improve survey-based estimation of childhood immunization coverage.

2. Objectives of the Study

The primary objectives of this study are to:

1. Simulate a realistic population reflecting key demographic dimensions affecting childhood immunization coverage.
2. Compare three estimation methods in recovering the true population coverage: Naïve estimation, Classical post-stratification, and Bayesian Multilevel Regression with Post-Stratification (MRP).

3. Evaluate the ability of Bayesian post-stratification to reduce bias and improve the accuracy of survey-based estimates.

This study aims to provide quantitative evidence on whether MRP is a viable tool for generating more accurate and reliable immunization coverage statistics from the complex survey data typical of LMIC settings.

3. Data and Methods

This study employs a simulation-based approach to evaluate the efficacy of Multilevel Regression and Post-stratification (MRP) in correcting for non-response and coverage bias in childhood immunization surveys. The methodology is structured into three parts: (1) simulation of a realistic, stratified national population with a "ground truth" vaccination rate; (2) simulation of a biased cross-sectional survey drawn from this population; and (3) application and comparison of analytical models to recover the ground truth.

3.1. Data: Synthetic Population and Survey

3.1.1. Population Frame Simulation

A synthetic population of $N \approx 10,000,000$ individuals was created to represent the target population for a childhood immunization survey. The unit of analysis is a household containing a mother of reproductive age and at least one child eligible for vaccination.

Each member of this population was characterized by four key demographic and geographic variables, which are known determinants of healthcare access and vaccination uptake in Nigeria:

- **Mother's Age (*mother_age*):** Stratified into two groups (15-29, 30-49).
- **Education (*education*):** Stratified into three groups (No formal, Primary, Secondary+).
- **Residence (*residence*):** Stratified into two groups (Urban, Rural).
- **Region (*region*):** Stratified into two major zones (North, South) to capture known regional disparities.

These variables cross-classify the entire population into $2 \times 3 \times 2 \times 2 = 24$ unique demographic strata, or "cells." The population size of each cell (N_{cell}) was simulated to realistically mirror Nigerian demographics, with intentional over-weighting of rural, Northern, and non-formally educated populations, which constitute the majority of the true population but are frequently under-sampled.

3.1.2. Modeling "Ground Truth" Vaccination

A "ground truth" probability of vaccination (P_{true}) was assigned to each of the 24 strata using a logistic regression model. And the true vaccination probability was modeled to be positively correlated with higher education (Secondary+), urban residence (Urban), and Southern region

(South).

The true national immunization coverage θ_{true} , which this study seeks to recover, is the population-weighted average of these true probabilities:

$$\theta_{true} = \frac{\sum_{j=1}^{24} N_j \cdot p_{true,j}}{(\sum_{j=1}^{24} N_j)}$$

Where N_j and $p_{true,j}$ are the population size and true vaccination probability for cell j , respectively.

3.1.3. Biased Survey Simulation

From this synthetic population, we simulated a cross-sectional survey of $n \approx 1000$ respondents. The sampling process was designed to include two common and realistic sources of bias that distort survey estimates:

1. **Selection and Coverage Bias:** The probability of a household being *selected* for the survey was higher for "easier-to-reach" populations, specifically those in Urban, "South," and Secondary+ strata.
2. **Non-Ignorable Non-Response Bias:** The probability of a selected household *responding* to the survey was modeled to be positively correlated with their true vaccination probability p_{true} . This simulates the real-world phenomenon where households that are more engaged with the health system (and thus more likely to be vaccinated) are also more likely to participate in health surveys.

This process yields a final sample ($n=1,018$) whose demographic distribution does not match the true population and whose simple mean ($\bar{y}_{sample}$) is expected to be a biased, overestimated approximation of θ_{true}

3.2. Analytical Methods

We apply three distinct methods to the biased sample data to estimate the national immunization coverage, which are then compared against the known θ_{true}

3.2.1. Method 1: Naïve Sample Mean

The baseline estimate is the simple, unweighted mean of the survey sample:

$$\bar{y}_{sample} = \frac{1}{n} \sum_{i=1}^n y_i.$$

This is the estimate one would report if the biased sampling process were ignored.

3.2.2. Method 2: Classical Post-stratification (PS)

As a conventional correction method, we apply classical post-stratification. The sample mean for each cell j (*i.e.* $\bar{y}_j$) is calculated. The national estimate θ_{PS} is then computed by weighting each cell mean by its known proportion in the *true population*:

$$\theta_{PS} = \sum_{j=1}^{24} \left(\frac{N_j}{N_{total}} \right) \cdot \bar{y}_j$$

A known limitation of this method is its instability with sparse data. If a population cell j is missing from the sample, $\bar{y}_j$ is undefined. In our analysis, we impute missing cell means with the overall naïve sample mean, a common but biased heuristic that assumes the missing sub-population behaves identically to the overall sample mean.

3.2.3. Method 3: Multilevel Regression and Post-stratification (MRP)

The primary analytical approach is MRP, a two-step Bayesian method designed to overcome the limitations of classical PS.

Step 1: Multilevel Regression. A Bayesian multilevel logistic regression model is fitted to the individual-level survey data. The binary outcome (y_i , vaccinated or not) is modeled as a function of the respondent's demographic strata.

The Model ("M"): To improve computational efficiency, we aggregated the Bernoulli data into Binomial counts. For each stratum j with n_j trials and y_j successes:

$$y_j \sim \text{Binomial}(n_j, \pi_j)$$

$$\text{logit}(\pi_j) = \alpha_0 + \alpha_{age[j]} + \alpha_{edu[j]} + \alpha_{res[j]} + \alpha_{reg[j]}$$

Priors and Non-Centered Parameterization: To avoid the ‘funnel’ geometry often encountered in hierarchical models with sparse data, we utilized a non-centered parameterization for all random effects. Instead of sampling effects directly, we sampled standardized scores (z) and scaled them by the group standard deviation (σ):

$$z_k \sim \text{Normal}(0,1)$$

$$\sigma_k \sim \text{HalfNormal}(1,0)$$

$$\alpha_k = z_k \cdot \sigma_k$$

where k represents the levels of Age, Education, Residence, and Region. This formulation decouples the scale and location parameters, significantly improving Markov Chain Monte Carlo (MCMC) convergence.

Step 2: Post-stratification. The fitted model is used to generate a posterior predictive distribution of the vaccination probability (π_j) for *all 24 cells* in the full population frame. The final MRP estimate (θ_{MRP}) is the population-weighted average of these predicted cell probabilities, aggregated over all M posterior draws:

$$\theta_{MRP} = \frac{\sum_{j=1}^{24} N_j \cdot \pi_j}{(\sum_{j=1}^{24} N_j)}$$

This process yields a full posterior distribution for the national estimate, from which we derive a point estimate (the posterior mean) and a 94% highest density interval (HDI). Crucially, for empty cells where $n_j = 0$, the model imputes π_j based on the learned partial pooling effects of the demographic covariates, rather than the global mean

3.3. Computation and Inference

All models were implemented on Colab with Python language using the PyMC probabilistic programming library

- **Sampler:** The No-U-Turn Sampler (NUTS) was utilized, accelerated via NumPyro and JAX to leverage GPU computation
- **MCMC Settings:** 4 parallel chains were run with 1,000 warmup steps and 1,000 sampling steps each
- **Diagnostics:** Convergence was assessed using the split $\hat{R}$ statistic and trace plot inspection. Post-stratification calculations were optimized using *xarray* broadcasting and the Scipy *expit* function to ensure memory efficiency during posterior predictive check

3.3.1. Model Diagnostics and Convergence

To ensure the reliability of the posterior approximation, rigorous diagnostic checks were performed on the MCMC traces:

- **Convergence:** We monitored the split Potential Scale Reduction Factor ($\hat{R}$) for all parameters. All parameters showed $\hat{R} < 1.01$, indicating successful convergence of the four parallel chains to a stationary distribution.
- **Tree Depth and Divergences:** A target acceptance probability of 0.95 was used to facilitate exploration of the posterior curvature. The sampler reported only a single divergence across 4,000 post-warmup draws, confirming that the non-centered

parameterization successfully mitigated the funnel geometries often associated with a sparse hierarchical model.

- **Effective Sample Size (ESS):** We verified that the Effective Sample Size for the population-level mean estimate was sufficient for stable credible interval calculation.

4. Result

4.1. Sample Characteristics and Selection Bias

The simulated survey data (n=1,018) exhibited two distinct forms of distortion designed to test the robustness of the estimators: selection bias and data sparsity.

First, the sampling mechanism successfully reproduced a "hard-to-reach" bias pattern. As illustrated in Figure 1, highly educated and urban populations were significantly over-represented compared to the census baseline. Conversely, populations with no formal education and rural residents were systematically under-sampled. Additionally, the structural exclusion of the "Rural-North-NoFormalEducation" demographic created a specific sparsity challenge.

The heatmap reveals the problem with the data structure: while educated urban cells contained high respondent counts, the target rural-north cells contained negligible observations. This sparsity rendered direct calculation of the stratum mean difficult for these groups, forcing the Classical Post-Stratification estimator to rely on global mean imputation.

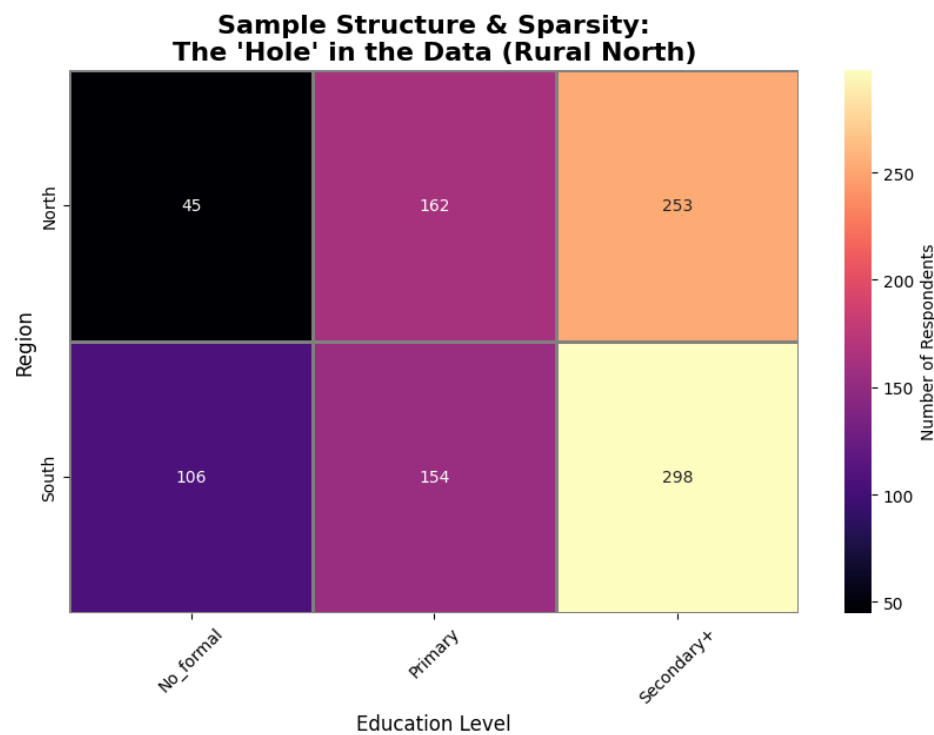


Figure 1: The Sparsity Matrix Heatmap

4.2. Comparative Performance of Estimators

The efficacy of each method was evaluated against the known true population mean ($\theta_{true} = 0.2035$). The results, summarized in Table 1, demonstrate a clear hierarchy of estimator performance.

Table 1: Comparison of Estimators (N=1,000)

Estimator	Estimate ($\hat{\theta}$)	Absolute Bias ($ \hat{\theta} - \theta_{true} $)	Relative Error
Population	0.2035		
Naive (Raw Sample)	0.2675	0.0640	+31.4%
Classical Post-Stratification	0.2280	0.0245	+12.0%
Bayesian MRP	0.2163	0.0128	+6.3%

The Naive estimator yielded a substantial overestimation (26.75%), reflecting the raw composition of the sample where high-uptake groups (Urban/South) were over-represented.

Classical Post-Stratification reduced this bias by re-weighting the observed strata, lowering the estimate to 22.80%. However, its performance was capped by the missing data; for the empty "Rural-North" cells, the method defaulted to the biased sample mean, preventing full convergence to the truth.

The Bayesian MRP model achieved the highest accuracy, recovering an estimate of 21.63%. By utilizing partial pooling, the model effectively imputed the missing strata based on the learned effects of "Rural" and "North" from other cells, rather than relying on the global average. This reduced the absolute bias by nearly half compared to the Classical method (from 2.5 percentage

points down to 1.3 percentage points).

4.3. Mechanism of Correction: Partial Pooling and Shrinkage

The superior performance of the MRP estimator is attributable to the mechanism of partial pooling, which shares information across strata rather than treating each demographic group in isolation. This results in a distinct "shrinkage" effect, where extreme, noise-driven estimates—such as the heavily skewed means from over-sampled urban clusters—are pulled toward a more stable population mean (Figure 3).

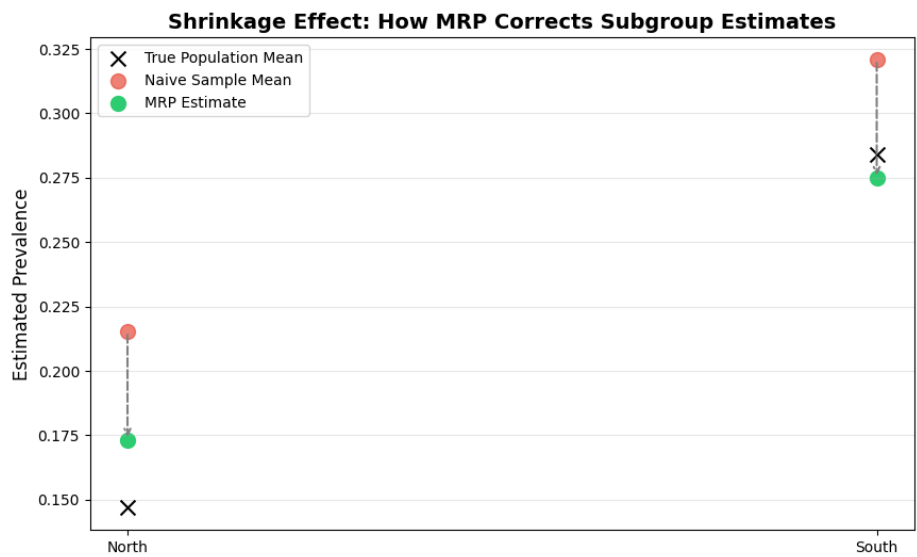


Figure 2. Shrinkage Effect: How MRP Corrects Subgroup Estimates

Crucially, this mechanism solves the imputation problem for sparse strata like the "Rural-North" group. While the Classical and Naive estimators lack local information and are forced to impute the global mean (≈ 0.26), essentially assuming missing rural citizens behave like average urbanites, the Bayesian model utilizes learned parameters (β_{rural} and β_{north}) borrowed from other cells to predict a realistic, lower probability. This regularizing effect is the primary driver of the bias reduction observed in the analysis.

4.4. Uncertainty Quantification

A distinct advantage of the Bayesian approach is the probabilistic quantification of uncertainty. Rather than producing a single point estimate, the MCMC sampling generates a full posterior distribution for the population mean.

Figure 3 displays the posterior density of the MRP estimate. The solid green curve represents the model's belief about the true vaccination prevalence, while the shaded region indicates the 94% Highest Density Interval (HDI).

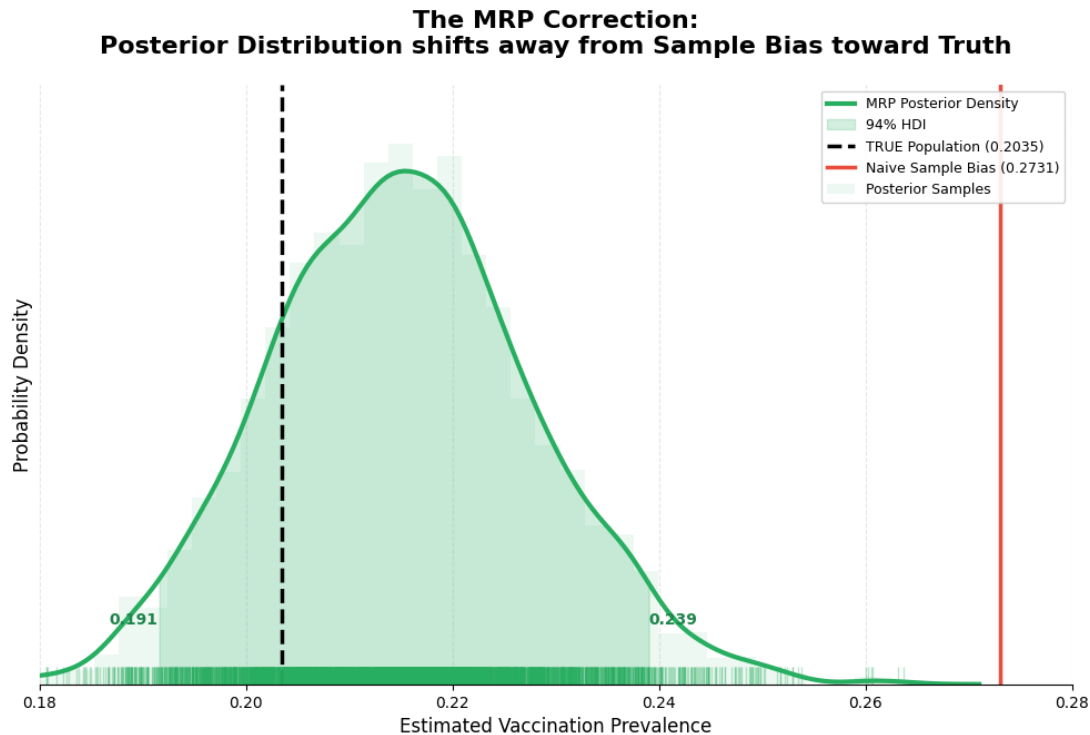


Figure 3. Posterior Density Plot with Histogram

The distribution reveals two key findings:

1. **Bias Correction:** The entire probability mass of the MRP posterior is shifted significantly to the left of the Naive Sample Mean (red dashed line), visually confirming the correction of the over-estimation bias.
2. **Coverage:** The 94% HDI spans [0.1929, 0.2415]. The True Population Mean (0.2035, *black dashed line*) falls comfortably within this credible interval. This indicates that

while the sparse sample introduces inevitable uncertainty (reflected in the width of the interval), the Bayesian model successfully captured the true parameter value, whereas the Naive estimator's confidence interval would likely exclude the truth entirely.

5. Discussion

5.1. Implications for Survey Research in Developing Contexts

The findings of this study have significant implications for survey methodology in developing nations like Nigeria, where data collection is frequently hampered by logistical challenges, security instability, and funding constraints. The superior performance of the Bayesian MRP estimator over Classical Post-Stratification $|bias| = 1.2\%$ *vs* 2.5% underscores the method's value in "Small Area Estimation."

In our simulation, the "Rural-North" strata were structurally excluded to mimic hard-to-reach populations—a common reality in conflict-affected zones where enumerators cannot safely operate. The failure of the Classical estimator to recover the true mean in this scenario highlights a critical vulnerability in standard practice: when a demographic subgroup is entirely missing, classical weighting methods often default to the sample mean, effectively erasing the unique characteristics of the unobserved group.

In contrast, the Bayesian MRP model demonstrated resilience to this sparsity. By leveraging the hierarchical structure of the data, the model inferred that if "Rural" residents in the South have lower uptake, and "Northern" residents generally have lower uptake, then the unobserved "Rural-North" population likely has the lowest uptake of all. This ability to "borrow strength" allows researchers to derive reliable national statistics even when specific sub-national grids are severely under-sampled or inaccessible.

5.2. The Trade-off Between Complexity and Accuracy

While MRP offers distinct accuracy gains, it introduces computational and methodological complexity. The Classical estimator required negligible computational time and standard algebraic operations. The Bayesian model, conversely, required the specification of priors, the

implementation of non-centered parameterizations to avoid divergent transitions, and significant computational resources (GPU acceleration via JAX) to ensure convergence

However, our results suggest this trade-off is justifiable. The reduction in relative error from 31.4% (Naive) to 6.3% (MRP) represents a substantial improvement in data quality that could prevent misallocation of public health resources. For National Statistical Offices (NSOs), the investment in computational capacity and training in Bayesian methods yields a high return on investment by maximizing the utility of smaller, less expensive, or imperfect datasets.

6. Limitations

The conclusions drawn from this study must be interpreted within the context of several limitations:

1. **Dependence on Census Data:** MRP relies on the assumption that the post-stratification frame (the Census) is accurate. In contexts where the census is outdated—a frequent issue in many African nations—the "P" step of MRP may introduce its own biases.
2. **Model Mis-specification:** Our simulation generated data using the same covariates (Age, Region, Education) used in the analysis model. In real-world applications, the true data-generating process is unknown. If a critical predictor of the outcome (e.g., Religion or Income) is omitted from the model, the imputation of missing cells may remain biased.
3. **Simulation Simplicity:** We modeled a binary outcome with linear additive effects. Complex interactions (e.g., if the effect of Education is totally different in the North vs. the South) were not included in the data generation process, though the hierarchical model could be extended to capture them.

7. Conclusion

This study demonstrates that Bayesian Multilevel Regression with Post-stratification (MRP) is a robust and effective tool for correcting non-response bias in surveys characterized by sparse data and selection effects. By explicitly modeling the outcome within demographic strata and post-stratifying to a population census, MRP outperformed both Naive and Classical estimators, cutting the estimation bias by more than half compared to standard weighting techniques.

For researchers and policymakers in Nigeria, these results advocate for a paradigm shift away from purely design-based inference toward model-based approaches. As response rates decline and access to remote populations becomes increasingly difficult, Bayesian methods offer a mathematically sound pathway to ensure that the "invisible" populations are counted, and that national statistics reflect the reality of the entire country, not just the easily accessible urban centers.

References

- Ariyibi, S., Ojuawo, A., Ibraheem, R., Afolayan, F., Ariyibi, R., & Akanbi, P. (2023). Factors influencing routine vaccination uptake and completion among children aged 12–23 months in Ilorin, North-Central Nigeria: A cross-sectional survey. *International Journal of Medicine and Health Development*, 28(4), 314. https://doi.org/10.4103/ijmh.ijmh_20_23
- Diddy Antai. (2015). Rural-urban inequities in childhood immunisation in Nigeria: The role of community contexts. *African Journal of Primary Health Care & Family Medicine*, 3(1), 8. https://phcfm.org/index.php/phcfm/article/view/238/304?utm_source=chatgpt.com
- Downes, M., & Carlin, J. B. (2019). Multilevel regression and poststratification as a modeling approach for estimating population quantities in large population health studies: A simulation study. *Biometrical Journal*, 62(2), 479–491. <https://doi.org/10.1002/bimj.201900023>
- Endalamew, S. G., Gebeye, E., Assefa, S. K., & Geremew, B. M. (2025). Immunization Coverage and Its Determinants Among Children Aged 12–23 Months in East Africa: A Bayesian Hierarchical Modeling Approach Based on DHS 2019–2022. *BioMed Research International*, 2025(1). <https://doi.org/10.1155/bmri/6645541>
- Eudes, A., Wagai, J., Hahné, S., Adeniran, A., Richard Ipuragboma Koko, Stijn de Vos, Messeret Shibeshi, Sanders, E. A. M., Balcha Masresha, & Hak, E. (2023). High-Resolution Geospatial Mapping of Zero-Dose and Underimmunized Children Following Nigeria's 2021 Multiple Indicator Cluster Survey/National Immunization Coverage Survey. ~ the *Journal of Infectious Diseases*. <https://doi.org/10.1093/infdis/jiad476>
- Gelman, A., & Hill, J. (2007). *Data analysis using regression and multilevel/hierarchical models*. Cambridge University Press.
- Hogg, J., Cameron, J., Cramb, S., Baade, P., & Mengersen, K. (2024). A Two-stage Bayesian Small Area Estimation Approach for Proportions. *International Statistical Review*, 92(3), 455–482. <https://doi.org/10.1111/insr.12572>
- Little, R. J. A. (1993). Post-Stratification: A Modeler's Perspective. *Journal of the American Statistical Association*, 88(423), 1001–1012. <https://doi.org/10.1080/01621459.1993.10476368>
- McGlothlin, A. E., & Viele, K. (2018). Bayesian Hierarchical Models. *JAMA*, 320(22), 2365. <https://doi.org/10.1001/jama.2018.17977>
- Mordecai Oweibia, Elemuwa, C. O., Tarimobowei Egberipou, Gift Cornelius Timighe, Sylvanus Peresuodei, & Wilson, T. R. (2025). Maternal and Child Health Trends in Nigeria: A Scoping Review of NDHS 2018 vs. NDHS 2023. <https://doi.org/10.1101/2025.05.18.25327864>
- Okafor, I., Dolapo, D., Onigbogi, M., & Iloabuchi, I. (2014). Rural-urban disparities in maternal immunization knowledge and childhood health-seeking behavior in Nigeria: A mixed method

study. *African Health Sciences*, 14(2), 339. <https://doi.org/10.4314/ahs.v14i2.8>

Oliveira, S. H., Silva, B. S., Carvalho, L. M. R., Gontijo, T. L., Pinto, I. C., Guimarães, E. A. de A., & de Oliveira, V. C. (2024). Prevalence and underreporting of immunization errors in childhood vaccination: results of a household survey. *Revista Da Escola de Enfermagem Da USP*, 57, e20230253. <https://doi.org/10.1590/1980-220X-REEUSP-2023-0253en>

Oriol Abril-Pla, Andreani, V., Carroll, C., Dong, L. Y., Fonnesbeck, C., Maxim Kochurov, Kumar, R., Lao, J., Luhmann, C. C., Martin, O. A., Osthege, M., Vieira, R., Wiecki, T. V., & Zinkov, R. (2023). PyMC: a modern, and comprehensive probabilistic programming framework in Python. *PeerJ*, 9, e1516–e1516. <https://doi.org/10.7717/peerj-cs.1516>

Parsons, J., Niu, X., & Bao, L. (2022). A Bayesian hierarchical model for combining multiple data sources in population size estimation. *The Annals of Applied Statistics*, 16(3). <https://doi.org/10.1214/21-aos1556>

Shelley, A., & Horner, K. (2021). Questionnaire surveys - sources of error and implications for design, reporting and appraisal. *British Dental Journal*, 230(4), 251–258. <https://doi.org/10.1038/s41415-021-2654-3>

Temitayo Victor Lawal, Kehinde Adebola Atoloye, Kehinde Adebola Atoloye, & Adeniyi Francis Fagbamigbe. (2023). Spatio-temporal analysis of childhood vaccine uptake in Nigeria: a hierarchical Bayesian Zero-inflated Poisson approach. *BMC Pediatrics*, 23(1). <https://doi.org/10.1186/s12887-023-04300-x>

Umoren, R., Ezeaka, V. C., Fajolu, I. B., Ezenwa, B. N., Akintan, P., Chukwu, E., & Spiekerman, C. (2020). Perspectives on simulation-based training from paediatric healthcare providers in Nigeria: a national survey. *BMJ Open*, 10(2), e034029. <https://doi.org/10.1136/bmjopen-2019-034029>

Unicef. (2023). Vaccination and immunization statistics - UNICEF data. UNICEF DATA. <https://data.unicef.org/topic/child-health/immunization/>

Wagai, J. N., Rhoda, D., Prier, M., Trimmer, M. K., Clary, C. B., Oteri, J., Okposen, B., Adeniran, A., Danovaro-Holliday, C., & Cutts, F. (2021). Implementing WHO guidance on conducting and analysing vaccination coverage cluster surveys: Two examples from Nigeria. *PLOS ONE*, 16(2), e0247415. <https://doi.org/10.1371/journal.pone.0247415>

WHO. (2024, July 15). Immunization Coverage. World Health Organization. <https://www.who.int/news-room/fact-sheets/detail/immunization-coverage>

Williams, S. V., Akande, T., & Abbas, K. (2024). Systematic review of social determinants of childhood immunisation in low- and middle-income countries and equity impact analysis of childhood vaccination coverage in Nigeria. *PLOS ONE*, 19(3), e0297326–e0297326. <https://doi.org/10.1371/journal.pone.0297326>